

Unit-06

Computer Simulation, Animation and Virtual Reality

6.1 Introduction to Computer Animation

Computer animation is the process of generating animated visuals through computer graphics, where still images are rapidly displayed in a sequence to create the illusion of continuous motion. Each frame in the sequence shows a slight progression or change in the object's position, shape, or appearance, which, when played at a specific speed (frames per second), results in smooth and realistic movement.

6.3 Introduction to Simulation

Simulation refers to the process of creating a virtual representation or model of a real-world system, process, or phenomenon to study its behaviour under various conditions. By imitating the functionality and interactions of the actual system, simulation allows for analysis, testing, and prediction without the risks, costs, or constraints of real-world experimentation.

Applications:

- Engineering (e.g., structural simulations)
- Medicine (e.g., virtual surgeries)
- Business (e.g., financial risk modelling)
- Computer Graphics (e.g., animations and visual effects)

6.4 Introduction to Virtual Reality

Virtual Reality (VR) is a technology that uses computer-generated simulations to create an artificial environment. It allows users to experience and interact with this environment as though it were real, by engaging their senses—primarily sight, sound, and touch. The interaction happens in real-time, enabling users to explore and manipulate virtual objects and spaces using specialized devices like VR headsets, controllers, gloves, or sensors.